Costate control for nonlinear system identification with applications to excitable systems

Thiago B. Burghi*, Giovanni Pugliese Carratelli, and Michael Everett Rule

***Abstract*— Gradient-based parameter estimation for nonlinear systems can suffer from instability of the costate (adjoint) dynamics that characterizes backpropagation through time. This is in particular a significant problem in realistic models of neural excitability, where positive feedback loops create ill-conditioned optimization landscapes with exploding and vanishing gradients. For this reason, large-scale artificial spiking neural networks usually omit key aspects of biological neural computation. This paper takes a first step towards resolving this issue by introducing Surrogate Gradients by Costate Control (SGCC). In this method, we design gradient regularizers by directly controlling the costate (adjoint) dynamics associated to a parameter estimation problem. We apply this idea to a wide class of conductance-based neuronal models. In this context, we design a proportional high-gain controller that provably eliminates exploding gradients, and a state-dependent controller that exploits the state structure of biological neurons. Numerical computation on the Hodgkin-Huxley neuron model validate our findings. Overall, SGCC provides a framework for estimating the parameters of nonlinear systems with a high degree of sensitivity to variations in inputs and model parameters.**

I. INTRODUCTION

Gradient-based system identification and machine learning methods scale well—but only when system components and their interconnections keep gradients numerically well-conditioned. However, many dynamical systems important in science and engineering have regions of state space where trajectories rapidly contract or expand. When estimating model parameters from data, such regions cause vanishing and exploding gradients in the cost function associated with the optimization problem [1], [2].

The problems of vanishing and exploding gradients are especially acute in spiking neural networks (SNNs), which have been promoted as an energy-efficient alternative to traditional inference models. To address these problems in the particular case of discrete-time networks of Leaky Integrate-and-Fire (LIF) neurons, [3] proposed the method of *surrogate gradients*. In this method, cost function gradients are replaced in the optimization algorithm by their so-called surrogates, which are obtained via ad-hoc modifications to the backward pass of gradient computation. This has been shown empirically to drastically improve the performance of integrate-and-fire SNNs [4], which has contributed to their rapid adoption in neuromorphic engineering [5].

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Existing surrogate gradient methods are not applicable to dynamical systems where localized instabilities emerge from differentiable model equations. This is the case for conductance-based models of neuronal excitability, where positive feedback loops can lead to bifurcations [6] and ultra-sensitivity [7], [8]. Motivated by this application, this work introduces Surrogate Gradients by Costate Control (SGCC), which formalize the idea of surrogate gradients for optimizing the parameters of continuous-time nonlinear systems with differentiable vector fields.

The main idea in this paper is to treat the costate (adjoint) system characterizing the backward pass of gradient computation in a dynamic parameter estimation problem as a system to be regulated with feedback control. We define exploding gradients in a Lyapunov-theoretic fashion [9], and propose the design of controllers that limit the expansion of the costates involved in gradient computation. We show that such controllers can be derived from contracting nonlinear observers [10], [11], and then apply our ideas to conductance-based models of neuronal excitability. We contrast two particular costate controller designs: a proportional high-gain controller that provably eliminates exploding gradients, and a heuristic controller that exploits a separation between states associated with positive and negative feedback [12].

The idea of modifying costate dynamics has previously been explored in engineering applications by [13], who externally stabilise the Hamiltonian system associated to an optimal control problem to design controllers for offshore wind turbines. Similarly, in [14] and references therein the authors externally stabilize an adjoint dynamics to efficiently simulate turbulent flow. To the authors' knowledge, controlled adjoints or costates have not been explored nonlinear system identification. We note that while the parameter estimation difficulties posed by "severely" nonlinear systems may be alleviated through more standard uses of contracting observers [10], [15], [16], the methodology presented in this paper is applicable to systems for which contracting observers are not known, systems for which noise precludes the use of high-gain designs, and systems where uniform contraction over the whole state space is not desirable or achievable. Importantly, our approach opens new routes to principled design of heuristic estimators that draw on the established repertoire of controller design methods.

The paper is organized as follows: in Section II we introduce the parameter estimation and exploding gradient problems, as well as the class of conductance-based models; in Section III we present our main theoretical results, and numerically evaluate our costate controllers using the

Hodgkin-Huxley conductance model.

Notation: We use $\mathbb{R}_>$, $\mathbb{R}_\geq$ and $\mathbb{R}$ to denote the set of positive real numbers, non-negative real numbers, and real numbers, respectively. Similarly $\mathbb{Z}_\geq$ and $\mathbb{Z}_>$ denote the set of non-negative integer numbers and positive integer numbers, respectively. We use ∇ to denote the gradient of a function, and $\odot$ to denote the Hadamard product.

II. Preliminaries

A. Problem statement

We consider a nonlinear system satisfying

$$\begin{aligned} \dot{x} &= f_\theta(t,x), \qquad x(0)=\xi \\ y &= Cx \end{aligned} \tag{1}$$

where $x(t) \in X \subseteq \mathbb{R}^n$ is a state vector, $y(t) \in \mathbb{R}^m$ is an output vector, and $C \in \mathbb{R}^{m\times n}$ is a given readout matrix. The function f_θ is parametrized by $\theta \in \Theta \subseteq \mathbb{R}^d$, and is assumed to be continuously differentiable in θ as well as x. We assume that for any $(\theta,\xi) \in \Theta \times X$, the system (1) has a unique solution $x(t\,;\theta,\xi) \in X$, defined for all $t \geq 0$. We further assume that X is a compact set.

We seek to make use of (1) to estimate the dynamics underlying an input-output dataset $\{u^*(t), y^*(t)\}_{t\in[0,T]}$, where $u^*(t)$ is a known input, $y^*(t)$ is a known target output, and T is the time horizon of the dataset. The time-dependence of f in (1) is due to the known input $u^*(t)$, that is, $f_\theta(t,x(t)) := \tilde{f}_\theta(x(t),u^*(t))$ for some appropriately defined $\tilde{f}_\theta$.

We consider the problem defined below.

***Problem** 1:* Consider the system defined in (1). We seek to solve the following problem

$$\min_{\theta,\xi} \mathcal{V}(\theta,\xi) \tag{2}$$

where

$$\mathcal{V}(\theta,\xi) = \frac{1}{2T}\int_0^T \|y^*(t) - Cx(t\,;\theta,\xi)\|^2 dt + \rho(\theta) \tag{3}$$

The term $\rho(\theta)$ is a regularizing term, and T is the duration of the dataset[1].

Problem 1 is frequently addressed via first-order methods such as gradient descent and its variations. This requires the computation of the cost function gradients $\nabla_\theta \mathcal{V}(\theta,\xi)$ and $\nabla_\xi \mathcal{V}(\theta,\xi)$. Using variational arguments, the gradients can be written as

$$\nabla_\theta \mathcal{V}(\theta,\xi) = \mu(0;\theta,\xi), \quad \nabla_\xi \mathcal{V}(\theta,\xi) = \lambda(0;\theta,\xi) \tag{4}$$

where $\lambda(t;\theta,\xi)$ and $\mu(t;\theta,\xi)$, defined for $t \in [0,T]$, are solutions of the costate equations

$$\begin{aligned} \dot{\lambda} &= -\nabla_x f_\theta(t,x(t))\lambda + T^{-1}C^\intercal(y^*(t) - Cx(t)) \\ \dot{\mu} &= -\nabla_\theta f_\theta(t,x(t))\lambda \end{aligned} \tag{5}$$

subject to the terminal conditions

$$\lambda(T) = 0, \quad \mu(T) = \nabla_\theta \rho(\theta) \tag{6}$$

[1]It should be noted that $x(t\,;\,\theta,\xi)$ depends on θ but also on the initial state ξ since the state of the data is not accessible.

We give a self-contained derivation of (4)-(6) in Appendix B. Note that in order to solve the costate equations (5)-(6), one must first solve the state equations (1). We also note that the solution (5) yields directly the gradient $\nabla_\theta \mathcal{V}$ and via numerical integration the costs $\mathcal{V}$ defined in (4). This process is sometimes known as *back propagation through time*.

B. Exploding and Vanishing gradients

Exploding and vanishing gradients can significantly affect algorithms based on gradient descent. These two problems can be explained in terms of the stability of the (linear time-varying) costate system (5). Specifically, when viewed backwards in time, the solutions of (5) which contract too quickly towards the origin represent vanishing gradients. When solutions instead grow too rapidly in magnitude, the gradients are said to explode. In this paper we focus on the problem of exploding gradients, which we define as follows:

***Definition** 1:* Problem 1 has exploding gradients if there exist no uniformly positive definite bounded matrices $P(t)$ and $Q(t)$ such that the inequality

$$\nabla_x f_\theta(t,x(t;\theta,\xi))^\intercal P + P\nabla_x f_\theta(t,x(t;\theta,\xi)) - \dot{P} \preceq -Q \tag{7}$$

is satisfied for all $(\theta,\xi) \in \Theta \times X$ and all $t \in [0,T]$.

Notice that exploding gradients preclude backwards-in-time input-to-state stability of the λ dynamics in (5) [9].

C. Conductance-based models of neuronal excitability

We apply our results to a wide class of conductance-based neuronal models. In such models, a neuronal membrane voltage $v(t) \in \mathbb{R}$ evolves according to

$$c\dot{v} = \sum_{i=1}^{d-1} \theta_i p_i^{r_i} n_i^{s_i}(\nu_i - v) + \theta_0(\nu_0 - v) + u^*(t) \tag{8a}$$

$$\dot{p} = \Lambda_+(v)(-p + \sigma_+(v)) \tag{8b}$$

$$\dot{n} = \Lambda_-(v)(-n + \sigma_-(v)) \tag{8c}$$

$$y = v \tag{8d}$$

where $c \in \mathbb{R}_>$ is a capacitance, $\theta_0 \in \mathbb{R}_>$ is a leak current conductance, $u^*(t) \in \mathbb{R}$ is a piecewise-continuous recorded input stimulus, and $(p,n) \in \mathbb{R}^{d-1} \times \mathbb{R}^{d-1}$ are internal states containing "gating" variables that can activate or inactivate membrane currents. Individual currents are given by

$$I_i = \theta_i p_i^{r_i} n_i^{s_i}(v - \nu_i),$$

where $\theta_i \in \mathbb{R}_>$ are maximal ionic conductances, $\nu_i \in \mathbb{R}$ are reversal potentials, and $r_i \in \mathbb{Z}_\geq$ and $s_i \in \mathbb{Z}_\geq$ are gating exponents[2]. The matrix-valued functions

$$\Lambda_\pm(v) = \mathrm{diag}\{\lambda_\pm^1(v), \ldots, \lambda_\pm^{d-1}(v)\} \tag{9}$$

and vector-valued functions

$$\sigma_\pm(v) = \mathrm{col}(\sigma_\pm^1(v), \ldots, \sigma_\pm^{d-1}(v)) \tag{10}$$

[2]We also consider $r_i = 0$ (or $s_i = 0$), in which case the corresponding p_i (or n_i) dynamics become unobservable and can be ignored. In the remainder of the paper, to simplify our notation, we keep all p_i and n_i dynamics in the formulation.

are continuously differentiable in v. While specific $\Lambda_\pm(v)$ and $\sigma_\pm(v)$ are found in Appendix A, we assume in general that these functions satisfy the following:

***Assumption** 1 (conductance-based properties):* We assume

i. Each element of $\Lambda_\pm(v)$ satisfies

$$0 < \underline{\lambda}^i_\pm \leq \lambda^i_\pm(v) \leq \overline{\lambda}^i_\pm, \quad \forall v \in \mathbb{R}$$

ii. Each element of $\sigma_\pm(v)$ is a sigmoidal function such that $\sigma^i_\pm(v) \in (0,1)$, $\frac{d}{dv}\sigma^i_\pm(v)$ is strictly sign-definite, and

$$\lim_{v\to\infty} \tfrac{d}{dv}\sigma^i_\pm(v) = \lim_{v\to-\infty} \tfrac{d}{dv}\sigma^i_\pm(v) = 0$$

iii. Let $\eta^i_\pm = (\sigma^i_\pm)^{-1}(0.5)$ denote the half-activation voltage of each sigmoid. Then

$$(\nu_i - \eta^i_+)\left.\frac{d\sigma^i_+}{dv}\right|_{v=\eta^i_+} > 0, \quad (\nu_i - \eta^i_-)\left.\frac{d\sigma^i_-}{dv}\right|_{v=\eta^i_-} < 0,$$

for all $i = 1, \ldots, d-1$.

It should be noted that Assumption 1.iii implies that, for slowly varying $v \approx \eta^i$, the gating variables p_i provide *positive* feedback to the v dynamics, while the gating variables n_i provide *negative* feedback to that dynamics (for an in-depth analysis of positive and negative feedback in excitable systems, see [12]). This notation differs from, but is equivalent to, that of standard formulations where gating variables are split between activation and inactivation variables [6].

We make use of Assumption 1 to establish the following:

***Lemma** 1:* Consider (8). For any $T > 0$, there exist two real numbers $v_{\min} \leq \min_i \nu_i$ and $v_{\max} \geq \max_i \nu_i$ such that if $\mathrm{col}(v(0), p(0), n(0)) \in X$, with

$$X := [v_{\min}, v_{\max}] \times [0,1]^{d-1} \times [0,1]^{d-1}, \tag{11}$$

then

$$\mathrm{col}(v(t), p(t), n(t)) \in X$$

for all $t \in [0, T]$.

Proof: We note that $u^*(t)$ is bounded on $[0, T]$ and the proof then follows directly from [16, Lemma 1]. ■

Hence by comparing (8) to (1), we have $x = \mathrm{col}(v, p, n)$, X as in Lemma 1, $\Theta = \mathbb{R}^d_>$, $n = 2d - 1$, and $C = 1$.

III. Results

We give our results providing effective design criteria to smooth the gradients discussed in Section II-A. In Section III-B we design costate controllers for conductance-based models. The controllers are then validated numerically in Section III-C.

A. Surrogate gradients by costate control

We address the issue of exploding gradients by introducing a control input in the costate equations (5). The controlled costate equations satisfy

$$\begin{aligned}\dot{\tilde{\lambda}} &= -[\nabla_x f_\theta(t,x) + K_\theta(t,x)]\tilde{\lambda} + \tfrac{1}{T}C^\intercal(y^*(t) - Cx)\\ \dot{\tilde{\mu}} &= -\nabla_\theta f_\theta(t,x)\tilde{\lambda}\end{aligned} \tag{12a}$$

subject to the terminal conditions

$$\tilde{\lambda}(T) = 0, \quad \tilde{\mu}(T) = \nabla_\theta \rho(\theta) \tag{12b}$$

where $K_\theta(t,x)$ is a controller to be designed. The variables $\tilde{\lambda}$ and $\tilde{\mu}$ are *surrogate costates* with respect to λ and μ. The solution to (12) yields

$$\nabla_\theta \tilde{\mathcal{V}}(\theta,\xi) = \tilde{\mu}(0,\theta,\xi), \quad \nabla_\xi \tilde{\mathcal{V}}(\theta,\xi) = \tilde{\lambda}(0,\theta,\xi)$$

which we call the *surrogate gradients* of Problem 1.

We employ surrogate gradients to replace the gradients when Problem 1 has exploding gradients. Specifically, we make use of K_θ in order to ensure that there exist uniformly positive definite bounded matrices $\tilde{P}(t)$ and $\tilde{Q}(t)$ such that

$$A_\theta(t, x(t;\theta,\xi))^\intercal \tilde{P} + \tilde{P}A_\theta(t, x(t;\theta,\xi)) - \dot{\tilde{P}} \preceq -\tilde{Q}, \tag{13a}$$

where

$$A_\theta(t,x) = \nabla_x f_\theta(t,x) + K_\theta(t,x), \tag{13b}$$

for all $(\theta,\xi) \in \Theta \times X$ and all $t \in [0,T]$.

Reassuringly, the surrogate gradients vanish at a global optimum of Problem 1 when a ground truth system generates the input-output data:

***Proposition** 1:* Let $y^*(t) = Cx^*(t)$ be the solution of (1) with $\theta = \theta^*$ and $\xi = \xi^*$. Consider the controlled costate equations (12) and assume that $\nabla_\theta \rho(\theta^*) = 0$. Then $\tilde{\lambda}(0,\theta^*,\xi^*) = \lambda(0,\theta^*,\xi^*) = 0$ and $\tilde{\mu}(0,\theta^*,\xi^*) = \mu(0,\theta^*,\xi^*) = 0$.

Proof: See Appendix C. ■

It is straightforward to find a $K_\theta(t,x)$ that stabilizes the $\lambda(t)$ in the sense of (12)-(13). One can take, for instance, $K_\theta(t,x) \equiv -\kappa I$ for large enough $\kappa > 0$ (as long as X is compact). Indeed, $K_\theta(t,x)$ can also be designed as follows:

***Proposition** 2:* Let

$$\dot{\hat{x}} = g_\theta(t, \hat{x}, y(t))$$

be an exponentially contracting observer[3] for (1), uniformly in $\theta \in \Theta$. Then

$$K_\theta(t,x) = \nabla_x g_\theta(t,x,y(t)) - \nabla_x f_\theta(t,x)$$

ensures that (13) is satisfied.

Proof: See Appendix D. ■

Other considerations apart from stability must be taken into account in the design of K_θ:

- While surrogate and true gradients coincide at the global optimum (Proposition 1), they might significantly differ at other points of the parameter landscape. This includes points where the true gradients may be informative.
- An overly generic design for K_θ can significantly filter the input $(y^*(t) - Cx(t))$ in (12), causing vanishing surrogate gradients that could stall gradient descent.

In this paper we focus on surrogate gradients for conductance-based models, where the design of costate controllers exploits the system architecture.

[3] See e.g. [10, Definition 1].

B. Surrogate gradients in conductance-based models

In this section, we seek to design surrogate gradient controllers that exploit the properties of conductance-based models introduced in Section II-C. We denote by

$$x = \mathrm{col}(v, p, n) \text{ and } \lambda = \mathrm{col}(\lambda_v, \lambda_p, \lambda_n)$$

the states of (8) and associated costates (5). Using this notation, we have

$$\nabla_x f_\theta(x) = \begin{bmatrix} G_{vv}(p,n) & G_{pv}(v,p)^\intercal & G_{nv}(v,n)^\intercal \\ G_{vp}(x) & -\Lambda_+(v) & 0 \\ G_{vn}(x) & 0 & -\Lambda_-(v) \end{bmatrix} \tag{14a}$$

where the matrix-valued entries are given by

$$G_{vv} = -\sum_{i=1}^{d-1} \theta_i p_i^{r_i} n_i^{s_i} - \theta_0 \tag{14b}$$

$$G_{vp} = \mathrm{col}(\ldots,\ r_i \theta_i p_i^{r_i - 1} n_i^{s_i} (\nu_i - v),\ \ldots) \tag{14c}$$

$$G_{vn} = \mathrm{col}(\ldots,\ s_i \theta_i p_i^{r_i} n_i^{s_i - 1} (\nu_i - v),\ \ldots) \tag{14d}$$

$$G_{pv} = \Lambda_+'(v)(-p + \sigma_+(v)) + \Lambda_+(v)\sigma_+'(v) \tag{14e}$$

$$G_{nv} = \Lambda_-'(v)(-n + \sigma_-(v)) + \Lambda_-(v)\sigma_-'(v) \tag{14f}$$

where we have used $'$ to denote d/dv. We now give the following results that provide controller design methods for conductance based neuronal models.

1) High-gain surrogate gradient controller: We start by observing that a particular high-gain proportional controller directly addresses the problem of exploding gradients:

***Proposition** 3:* Consider (8) and (12). Let

$$K_\theta(t,x) = K^{\mathrm{prop}} = -\gamma \begin{bmatrix} 1 & 0_{1\times(d-1)^2} \\ 0_{(d-1)^2\times 1} & 0_{(d-1)^2\times(d-1)^2} \end{bmatrix} \tag{15}$$

for some $\gamma > 0$. Then for any $T > 0$, there is a large enough $\gamma > 0$ such that (13) is satisfied with

$$\tilde{P} = I, \quad \tilde{Q} = \mathrm{diag}\{\theta_0, \underline{\lambda}_+^1, \ldots, \underline{\lambda}_+^{d-1}, \underline{\lambda}_-^1, \ldots, \underline{\lambda}_-^{d-1}\}$$

for all $(\theta, \xi) \in \Theta \times X$ and for all $t \in [0, T]$, where $\Theta = \mathbb{R}^d_>$ and X is given by (11).

Proof: See Appendix E ■

Notice the high-gain controller of Proposition 3 guarantees the absence of gradient explosion by exploiting the fact that the dynamics of p and n are exponentially contracting, when v is viewed as an input to those dynamics.

2) Local suppression of positive feedback: We now propose a heuristic for designing a controller K_θ that locally suppresses the effect of positive feedback in the gradient calculation for conductance-based models. We base this design on the following argument. For each element p_i in p, assume that $p_i(t) \approx \sigma_+^i(v(t))$ whenever $v(t) \approx \eta_+^i$. This frequently occurs when the voltage dynamics is slow, relative to the time constant $\lambda_+^i(v(t))^{-1}$, as $v(t)$ approaches the half-activation η_+^i of the sigmoid σ_+^i. In this case, by making use of Assumption 1, each element of J_{vp} has the same sign as the corresponding element of $J_{pv} \approx \Lambda_+(v)\sigma'(v)$, whenever $v(t) \approx \eta_+^i$ in (14). This establishes a temporally localized positive feedback mechanism (in the sense of monotone systems; see [17]) between λ_v and λ_p that contributes to the explosion of gradients.

To reduce the effects of positive feedback triggered by p, we propose a costate controller that exploits the partition of gating variables in conductance-based models. The proposed controller is given by

$$K_\theta^{\mathrm{gat}}(x) = -(1-\alpha) \begin{bmatrix} 0 & K_{pv}(v)^\intercal & 0 \\ K_{vp}(x) & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{16}$$

with

$$K_{vp}(x) = J_{vp}(x)$$
$$K_{pv}(x) = \Lambda_+(v)\sigma_+'(v)$$

and $\alpha \in (0,1)$ is a controller gain. Notice that the effect of the above controller is to decrease the destabilizing off-diagonal terms in $\nabla_x f_\theta(x) + K_\theta^{\mathrm{gat}}(x)$ by a factor of $\alpha < 1$. The above controller acts therefore locally in state space, avoiding substantial changes to the costate dynamics in an attempt to preserve useful information contained in the original gradients.

While a formal stability proof is deferred to future work, in the next section, we numerically evaluate this control scheme and compare it to the earlier proportional high-gain design.

C. Computational results

In order to illustrate the practical significance of our results, we consider Problem 1 when $y^* = v^*$ is the membrane voltage of the celebrated Hodgkin-Huxley (HH) model [18] (see Appendix A for a description of the dynamics). In terms of (8), the data-generating HH model is defined with

$$\theta^* = (\theta_0^*, \theta_1^*, \theta_2^*) \equiv (\theta_{\mathrm{L}}^*, \theta_{\mathrm{Na}}^*, \theta_{\mathrm{K}}^*)$$

where θ_{L}^*, θ_{Na}^*, and θ_{K}^* are the maximal conductances of leak, sodium, and potassium membrane currrents, respectively. In what follows we have $\rho(\theta) \equiv 0$.

1) Single spike: We first illustrate the effectiveness of surrogate gradients in a simplified setting. We consider the data given by u^* and $y^* = v^*$ shown in Fig. 1, which also shows the state vectors p^* and n^* (not used when solving the problem). We seek an estimate the of the unknown parameter $\theta_{\mathrm{Na}} \equiv \theta_1$, while fixing the initial condition $\xi \equiv \xi^*$ and the remaining parameters $\theta_0 \equiv \theta_{\mathrm{L}}^*$ and $\theta_2 \equiv \theta_{\mathrm{K}}^*$. This leads to a non-convex cost $\mathcal{V}(\theta_{\mathrm{Na}})$ displaying an exploding gradient at approximately $\theta_{\mathrm{Na}} = 107.8$, which is shown in Fig. 2 (top two panels). Fig. 2 shows also that the exploding gradient occurs due to the sudden appearance of an action potential in the output $v(t)$ of the estimator model. This phenomenon is ultimately related to the existence of a subcritical Hopf bifurcation in the Hodgkin-Huxley model, near the global optimum $\theta_{\mathrm{Na}}^* = 120.0$ (see *e.g.* [6]).

We now make use of our results to smooth the gradient and leave the global minimum of the problem unchanged. Fig. 2 shows the surrogate gradients $\nabla_{\theta_{\mathrm{Na}}} \tilde{\mathcal{V}}_{\mathrm{prop}}$ and $\nabla_{\theta_{\mathrm{Na}}} \tilde{\mathcal{V}}_{\mathrm{gat}}$ obtained respectively with the proportional controller K^{prop} from (15), with $\gamma = 4$, and the gating-based controller K^{gat} from (16), with $\alpha = 0.7$. In addition, we also show the

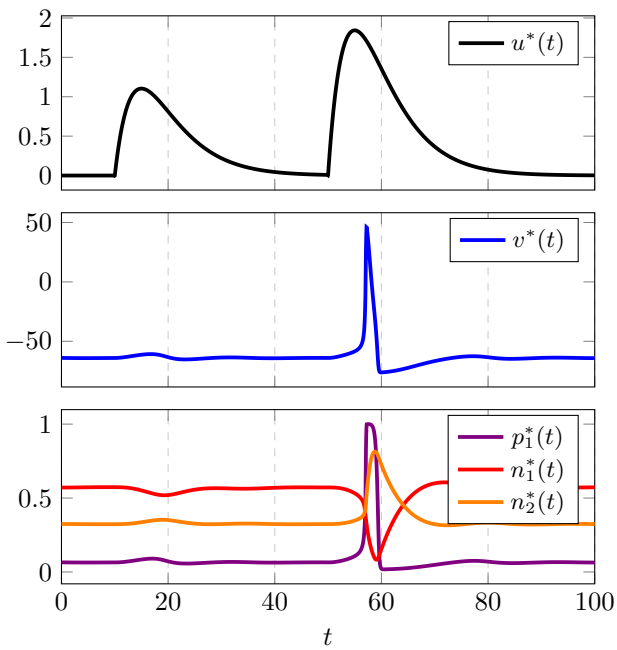

Fig. 1: Time evolution of system (8) using parameters of the Hodgkin-Huxley model, described in Appendix A. The variables $y^* = v^*$ and u^* constitute the data used to solve Problem 1; for illustrative purposes, we also show the gating variables $p_1^*(t)$, $n_1^*(t)$, and $n_2^*(t)$ corresponding to sodium activation, sodium inactvation, and potassium activation, respectively.

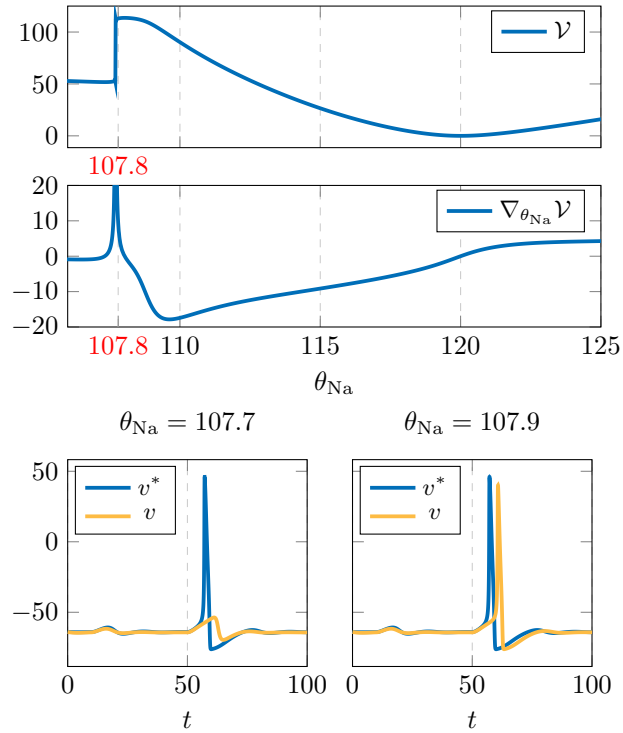

Fig. 2: **Top:** cost function $\mathcal{V}(\theta_{\mathrm{Na}})$ and corresponding gradient $\nabla_{\theta_{\mathrm{Na}}}\mathcal{V}(\theta_{\mathrm{Na}})$ associated to Problem 1, computed numerically using the data (v^*, u^*) shown in Fig. 1, for a Hodgkin-Huxley estimator model where θ_{Na} is a free variable and all other parameters are set to ground truth. The non-convex cost displays a jump at $\theta_{\mathrm{Na}} \approx 107.8$, causing an exploding gradient that could disrupt parameter estimation by gradient descent. Notice the ground truth value of θ_{Na} is given by $\theta^*_{\mathrm{Na}} = 120$. **Bottom:** inspection of model trajectories near $\theta_{\mathrm{Na}} \approx 107.8$ reveals the cause of the exploding gradient: an action potential appears as θ_{Na} is increased towards the global optimum. Specific values of model parameters are provided in Appendix A.

surrogate costs $\tilde{\mathcal{V}}_{\mathrm{prop}}$ and $\tilde{\mathcal{V}}_{\mathrm{gat}}$ obtained by integrating the surrogate gradients over θ_{Na}. It can be seen that, in agreement with Proposition 1, both surrogate gradient functions vanish at the global optimum $\theta_{\mathrm{Na}} = \theta^*_{\mathrm{Na}} = 120.0$ of the original problem (*cf* Fig. 1, top panel).

Fig. 3 shows that both costate control schemes presented in Section III-B eliminate the gradient explosion from the original optimization problem (*cf* Fig. 2). The gating-based costate controller is able to considerably smooth the gradient around the location of the original exploding gradient. This results in a cost function with no local minima other than the global one. For $\gamma = 4$ the proportional costate controller also smooths the original gradients. The surrogate cost function $\mathcal{V}^{\mathrm{prop}}$ retains however a local minimum near the location of the original gradient singularity. The bottom panel of Fig. 3 shows the surrogate gradients for the proportional design using other values of γ; it should be noted that high values of γ might not always be desirable since these can lead to uniform flattening of the gradient.

2) Spike train: In Fig. 4 we show the more realistic scenario where the data is made of a train of spikes (top two panels), triggered by a deterministic input mimicking pink noise (see Appendix A for simulation parameters). The original cost function, associated with Problem 1 and the Hodgkin-Huxley model with unknown θ_{Na}, is now considerably more challenging to optimize using standard gradient-based techniques. In particular, it possesses several local minima occurring near gradient singularities (middle panel).

Fig. 4 also shows the resulting surrogate cost functions obtained using the same K^{prop} and $K^{\mathrm{gat}}_{\theta}$ from the previous example, with $\gamma = 4$ and $\alpha = 0.7$ respectively. We observe similar effects to those of the simpler data presented before: the controllers effectively stabilise the costate dynamics, and yield a notably smoother cost function. In this example, $\tilde{V}_{\mathrm{prop}}$ does not have extra local minima, while $\tilde{V}_{\mathrm{gat}}$ possesses a region of small gradients for high values of θ_{gat}.

IV. CONCLUSIONS AND FUTURE WORK

In this paper we take a first step in using a control action to stabilize the costate dynamics resulting from a parameter estimation problem. We have introduced the idea of costate control in general terms, and applied it to a relevant class of nonlinear systems where exploding gradients resulting from instabilities of the costate dynamics are all but guaranteed to occur.

Future work will address methodological and computational developments of our results. In terms of methodology, future work will address stability guarantees for the gating-based controller, as well as adaptive strategies for the selection of γ and α when several parameters must be estimated. In terms of computation, we will exploit the methodology in the practical context of learning intrinsic and synaptic conductances in a spiking neural network.

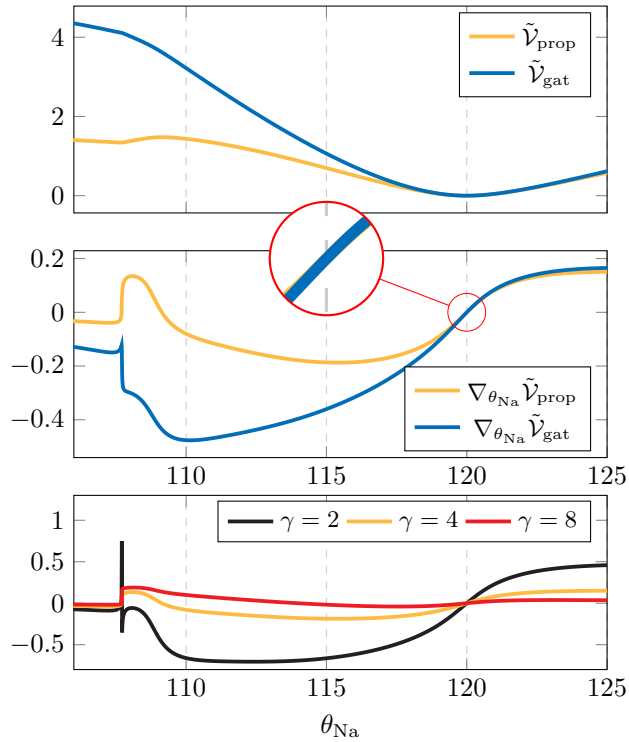

Fig. 3: **Middle:** surrogate gradients $\nabla_{\theta_{\text{Na}}}\tilde{\mathcal{V}}_{\text{prop}}$ obtained with the proportional controller K^{prop} from (15) with $\gamma = 4$, and $\nabla_{\theta_{\text{Na}}}\tilde{\mathcal{V}}_{\text{gat}}$ obtained with the gating-based controller K^{gat} from (16) with $\alpha = 0.7$. **Top:** surrogate costs $\tilde{\mathcal{V}}_{\text{prop}}$ and $\tilde{\mathcal{V}}_{\text{gat}}$ obtained by integrating the surrogate gradients over θ_{Na}. An arbitrary integration constant was added so that $\tilde{\mathcal{V}}_{\text{prop}}(\theta^*) = \tilde{\mathcal{V}}_{\text{gat}}(\theta^*) = 0$. **Bottom:** surrogate gradients $\nabla_{\theta_{\text{Na}}}\tilde{\mathcal{V}}_{\text{prop}}$ obtained with different values of the proportional gain γ.

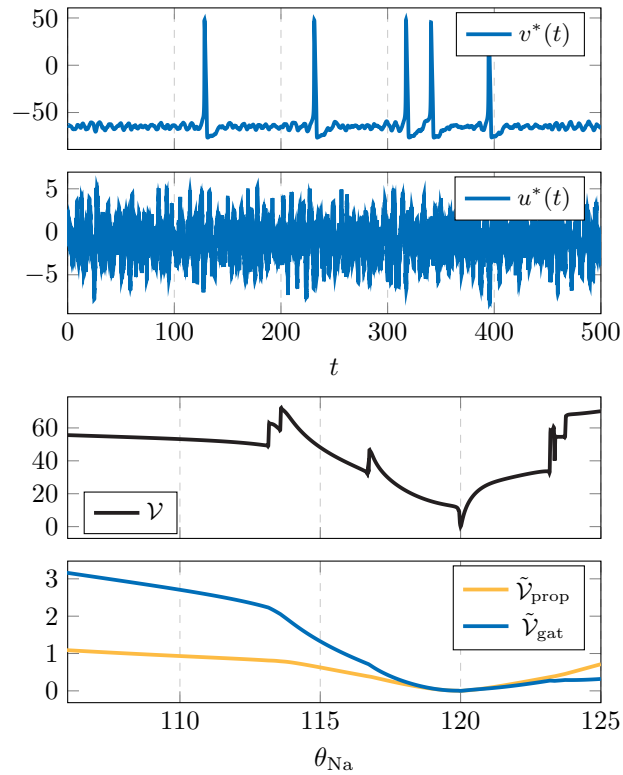

Fig. 4: **Top:** a realistic dataset containing a train of spikes v^* elicited by a pseudo-noisy current u^*; see Appendix A for simulation parameters. **Bottom:** Cost function of the original problem of estimating the sodium conductance θ_{Na} of the Hodgkin-Huxley model subject to the data above, and cost functions computed from the surrogate gradients obtained with K_{prop} and K_{gat}.

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APPENDIX

The code for the simulations is available at [19].

A. Simulation parameters

1) Hodgkin-Huxley model: We base our numerical results on the HH model from [6, Chapter 2]. The model contains a leak current (L), a sodium current (Na), and a potassium current (K). These currents are typically denoted by

$$\begin{aligned} I_{\text{Na}} &= \theta_{\text{Na}} m_{\text{Na}}^3 h_{\text{Na}}(v - \nu_{\text{Na}}) \\ I_{\text{K}} &= \theta_{\text{K}} m_{\text{K}}^4 (v - \nu_{\text{K}}) \\ I_{\text{K}} &= \theta_{\text{L}}(v - \nu_{\text{L}}) \end{aligned}$$

where m_{Na} and m_{K} are activation gating variables, and h_{Na} is an inactivation gating variable. In the context of this work, the HH model is mapped to (8) according to

$$\begin{aligned} \theta &= (\theta_0, \theta_1, \theta_2) \equiv (\theta_{\text{L}}, \theta_{\text{Na}}, \theta_{\text{K}}) \\ \nu &= (\nu_0, \nu_1, \nu_2) \equiv (\nu_{\text{L}}, \nu_{\text{Na}}, \nu_{\text{K}}) \end{aligned}$$

and

$$p = \begin{bmatrix} m_{\text{Na}} \end{bmatrix}, \quad n = \begin{bmatrix} h_{\text{Na}} \\ m_{\text{K}} \end{bmatrix} \tag{17}$$

which highlight the fast positive feedback role of sodium activation [12]. The specific values of θ_i^* and fixed ν_i used to generate the problem data v^* and u^* can be found in Table I. The nonlinearities $\Lambda_\pm(v)$ and $\sigma_\pm(v)$ in (9) and (10) are defined with the Gaussian-like function

$$\lambda_\pm^j(v) = a_\pm^j + b_\pm^j \exp\left(-\left(\frac{v - \xi_\pm^j}{\beta_\pm^j}\right)^2\right) \tag{18}$$

and the logistic function

$$\sigma_\pm^j(v) = \left(1 + \exp\left(-\frac{v - \eta_\pm^j}{\kappa_\pm^j}\right)\right)^{-1} \tag{19}$$

respectively. The specific values of the parameters appearing in the definitions above are summarised in Table II.

2) External inputs: The results in Fig. 1-Fig. 4 were obtained by first, numerically integrating the forward system ODEs and saving the data $v^*(t)$ and $u^*(t)$ at every $dt = 0.01$ ms; then numerically integrating the costate equations ODEs using linear interpolation between the saved data points. This was for every θ_{Na} between 106 and 125 with a step of $\Delta\theta_{\text{Na}} = 0.025$. In Fig. 1 the external current signal $u^*(t)$ is given by

$$u^*(t) = 15\,\Gamma(t - 10; \chi, \tau) + 20\,\Gamma(t - 50; \chi, \tau)$$

where

$$\Gamma(t;\chi,\tau) = t^{\chi-1}\exp(-t/\tau)$$

with $\chi = 2$ and $\tau = 5$. In Fig.4 we consider the external current to satisfy a deterministic approximation of pink noise given by the sum sinusoids with uniformly distributed random phases

$$u^*(t) = -1.0 + \frac{2}{\sqrt{K_{\text{sos}}}}\sum_{k=1}^{K_{\text{sos}}} a_k \sin(2\pi f_k t + \phi_k)$$

where $K_{\text{sos}} = 48$, $a_k = \frac{1}{\sqrt{f_k}}$, $f_k = \exp\left(\frac{\log f_k^{\text{edge}} + \log f_{k+1}^{\text{edge}}}{2}\right)$, $\log f_k^{\text{edge}} = \log(f_{\min}) + \frac{k-1}{K_{\text{sos}}}[\log(f_{\max}) - \log(f_{\min})]$, $f_{\min} = 0.1$, $f_{\max} = 10$, and $\phi_k \sim \mathcal{U}(0, 2\pi)$.

Parameter	Symbol	Value	Units
Capacitance	c	1.0	μF/cm^2
Leak Max. Conductance	$\theta_0 \equiv \theta_L$	0.3	mS/cm^2
Sodium Max. Conductance	$\theta_1 \equiv \theta_{Na}$	120.0	mS/cm^2
Potassium Max. Conductance	$\theta_2 \equiv \theta_K$	36.0	mS/cm^2
Leak Reversal Potential	$\nu_0 \equiv \nu_{\text{L}}$	-54.4	mV
Sodium Reversal Potential	$\nu_1 \equiv \nu_{\text{Na}}$	55.0	mV
Potassium Reversal Potential	$\nu_2 \equiv \nu_{\text{K}}$	-77.0	mV
Sodium activation exponent	r_1	3	–
Sodium inactivation exponent	s_1	1	–
Potassium activation exponent	r_2	4	–

TABLE I: Model Parameters and Constants.

Nonlinearities	η^j (mV)	κ^j	b^j (mV)	β^j	b^j	a^j
λ_+^1, σ_+^1 (p_1)	-40.0	9.0	-38.0	30.0	0.46	0.04
λ_-^1, σ_-^1 (n_1)	-62.0	-7.0	-67.0	20.0	7.4	1.2
λ_-^2, σ_-^2 (n_2)	-53.0	15.0	-79.0	50.0	4.7	1.1

TABLE II: Gating variable nonlinearity parameters.

B. Continuous-time gradients

To derive the continuous-time backpropagation equations, we first write the cost (3) in the equivalent form

$$\min_{\{\eta(t),x(t)\}_{t\geq 0}} \quad \frac{1}{2T}\int_0^T \|y^*(t) - Cx(t)\|^2 dt + \rho(\eta(T)) \tag{20a}$$

$$\text{subject to} \quad \begin{aligned} \dot{x} &= f_\eta(t, x(t)), \quad & x(0) &= \xi \\ \dot{\eta} &= 0, & \eta(0) &= \theta \end{aligned} \tag{20b}$$

where the trajectories $\eta(t)$ and $x(t)$ are free variables the problem for $t \in [0, T]$. The problem (20) is a dynamic programming problem with fixed final time $t = T$, free final states $\eta(T)$ and $x(T)$, and terminal cost $\rho(\eta(T))$ (see e.g. [20]). The Hamiltonian function of the problem is given by

$$H(x, \lambda, \mu) = \frac{1}{2T}\|y^*(t) - Cx(t)\|^2 + \lambda^\intercal f_\eta(t, x) + \mu^\intercal 0$$

where λ and μ are the costate vectors associated to the $\dot{x}$ and $\dot{\eta}$ dynamics, respectively. The optimal states are found by solving the Hamiltonian system, which is a set of ODEs given by the dynamical constraints (20b) alongside the costate dynamics

$$\begin{aligned} \dot{\lambda} = -\nabla_x H \quad &= -\nabla_x f_\eta(t, x)\lambda + \frac{1}{T}C^\intercal(y^\circ(t) - Cx) \\ \dot{\mu} = -\nabla_\eta H \quad &= -\nabla_\eta f_\eta(t, x)\lambda \end{aligned} \tag{21}$$

with boundary conditions $\lambda(T) = 0$ and $\mu(T) = \nabla_\eta \rho(\eta(T))$.

Let $\mathcal{V}(T, t, \eta_t, x_t)$ be the value function of the problem (20), that is, the cost-to-go associated to (20a) at time t and with states starting at $x(t) = x_t$ and $\eta(t) = \eta_t$. Since the final states are free and the final time is fixed, the costates yield the value function gradients

$$\begin{aligned} \lambda(t) &= \nabla_{x_t}\mathcal{V}(T, t, \eta(t), x(t)) \\ \mu(t) &= \nabla_{\eta_t}\mathcal{V}(T, t, \eta(t), x(t)) \end{aligned} \tag{22}$$

with $\eta(t)$ and $x(t)$ solutions to the Hamiltonian system. Since the gradients of the original cost function $\mathcal{V}(\theta, \xi)$ are given by $\nabla_\xi \mathcal{V}(\theta, \xi) = \nabla_{x_0}\mathcal{V}(T, 0, \theta, \xi)$ and $\nabla_\theta \mathcal{V}(\theta, \xi) = \nabla_{\eta_0}\mathcal{V}(T, 0, \theta, \xi)$, we obtain the desired gradients directly from (21) and (22).

C. Proof of Proposition 1

Consider the solution $x = x^*$ of (1) at the global optimum $\theta = \theta^*$ and $\xi = \xi^*$. Substituting x^* in the controlled costate equations (12) results in $C^\intercal(y^*(t) - Cx^*) = 0$. The surrogate dynamics thus become

$$\begin{aligned} \dot{\tilde{\lambda}}^* &= -[\nabla_x f_{\theta^*}(t, x^*) + K_\theta(t, x^*)]\tilde{\lambda}^* \\ \dot{\tilde{\mu}}^* &= -\nabla_\theta f_{\theta^*}(t, x)\tilde{\lambda}^* \end{aligned} \tag{23}$$

and hence the set

$$\{\tilde{\lambda}^* \in \mathbb{R}^n, \tilde{\mu}^* \in \mathbb{R}^d \mid \tilde{\lambda}^* = 0\} \tag{24}$$

is included in the set of equilibrium points of (23), with time understood to flow in the negative direction. From (12b) and the assumption that $\nabla_\theta \rho(\theta^*) = 0$, it follows that the surrogate gradients are obtained using the terminal conditions $\tilde{\lambda}(T) = 0$, $\tilde{\mu}(T) = 0$. Since the terminal costates belong to the set (24), we have $\tilde{\lambda}(t, \theta^*, \xi^*) = 0$ and $\tilde{\mu}(t, \theta^*, \xi^*) = 0$ for all $0 \leq t \leq T$, and the surrogate gradients vanish. By (4), the true costates also vanish at (θ^*, ξ^*) for $t = 0$.

D. Proof of Proposition 2

Since the observer is exponentially contracting, we can take $M = \tilde{P}^{-1}$ in Equation 11 of the converse theorem in [11, Section 3.5] to show that there is a uniformly positive definite $\tilde{P}$ and $\rho \in \mathbb{R}_>$ such that (13) is satisfied with $\tilde{Q} = \rho\tilde{P}$ for all $x \in \mathbb{R}^n$ (recall $\frac{d}{dt}\tilde{P}^{-1} = -\tilde{P}^{-1}\dot{\tilde{P}}\tilde{P}^{-1}$). In particular, (13) is satisfied along the trajectory $x = x(t; \theta, \xi)$.

E. Proof of Proposition 3

We show that for a sufficiently high $\gamma > 0$, the matrix

$$M := (\nabla_x f_\theta(x) + K^{\text{prop}})^\intercal \tilde{P} + \tilde{P}(\nabla_x f_\theta(x) + K^{\text{prop}}) + \tilde{Q}$$

is negative definite for all $x \in X$, which implies it is negative definite for all $t \in [0, T]$ when we substitute $x = x(t; \theta, \xi)$. To show this, write M as

$$M = \begin{bmatrix} M_1 & M_2^\intercal & M_3^\intercal \\ M_2 & M_4 & 0 \\ M_3 & 0 & M_5 \end{bmatrix}$$

and notice that by Assumption 1 and Lemma 1 we have

$$M_1 \leq -2\gamma - \theta_0,\ M_4 \preceq -\operatorname{diag}\{\underline{\lambda}_+^i\},\ M_5 \preceq -\operatorname{diag}\{\underline{\lambda}_-^i\},$$

Since all terms of $\nabla_x f_\theta(t, x)$ given by (14) are bounded on the compact set X, the same applies to M_2 and M_3 above. Hence we can use Schur's complement to show that for sufficiently large $\gamma > 0$, M is negative definite on X.

References

[1] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," in *Proceedings of the 30th International Conference on Machine Learning*, ser. ICML'13, vol. 28, Jun. 2013, pp. 1310–1318.

[2] A. H. Ribeiro, K. Tiels, J. Umenberger, T. B. Schön, and L. A. Aguirre, "On the smoothness of nonlinear system identification," *Automatica*, vol. 121, p. 109158, Nov. 2020.

[3] F. Zenke and S. Ganguli, "Superspike: Supervised learning in multilayer spiking neural networks," *Neural computation*, vol. 30, no. 6, pp. 1514–1541, 2018.

[4] F. Zenke and T. P. Vogels, "The remarkable robustness of surrogate gradient learning for instilling complex function in spiking neural networks," *Neural computation*, vol. 33, no. 4, pp. 899–925, 2021.

[5] B. Cramer, S. Billaudelle, S. Kanya, A. Leibfried, A. Grübl, V. Karasenko, C. Pehle, K. Schreiber, Y. Stradmann, J. Weis, J. Schemmel, and F. Zenke, "Surrogate gradients for analog neuromorphic computing," *Proceedings of the National Academy of Sciences*, vol. 119, no. 4, p. e2109194119, Jan. 2022.

[6] E. M. Izhikevich, *Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting*. The MIT Press, Jul. 2006.

[7] G. Drion, T. O'Leary, J. Dethier, A. Franci, and R. Sepulchre, "Neuronal behaviors: A control perspective," in *54th IEEE Conference on Decision and Control*, Dec. 2015, pp. 1923–1944.

[8] A. Franci, G. Drion, and R. Sepulchre, "The sensitivity function of excitable feedback systems," in *2019 IEEE 58th Conference on Decision and Control (CDC)*, Dec. 2019, pp. 4723–4728.

[9] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

[10] I. R. Manchester, "Contracting Nonlinear Observers: Convex Optimization and Learning from Data," in *2018 Annual American Control Conference (ACC)*, Jun. 2018, pp. 1873–1880.

[11] W. Lohmiller and J.-J. E. Slotine, "On Contraction Analysis for Nonlinear Systems," *Automatica*, vol. 34, no. 6, pp. 683–696, Jun. 1998.

[12] A. Franci, G. Drion, V. Seutin, and R. Sepulchre, "A Balance Equation Determines a Switch in Neuronal Excitability," *PLOS Computational Biology*, vol. 9, no. 5, p. e1003040, May 2013.

[13] T. Mylvaganam, M. Sassano, and A. Astolfi, "Nonlinear optimal control of a ballast-stabilized floating wind turbine via externally stabilised Hamiltonian dynamics," in *2021 60th IEEE Conference on Decision and Control (CDC)*, Dec. 2021, pp. 2428–2433.

[14] A. Garai and S. M. Murman, "Stabilization of the Adjoint for Turbulent Flows," *AIAA Journal*, vol. 59, no. 6, pp. 2001–2013, Jun. 2021.

[15] H. D. I. Abarbanel, D. R. Creveling, and J. M. Jeanne, "Estimation of parameters in nonlinear systems using balanced synchronization," *Physical Review E*, vol. 77, no. 1, Jan. 2008.

[16] T. B. Burghi and R. Sepulchre, "Adaptive observers for biophysical neuronal circuits," *IEEE Transactions on Automatic Control*, vol. 69, no. 8, pp. 5020–5033, 2024.

[17] D. Angeli, J. E. Ferrell, and E. D. Sontag, "Detection of multistability, bifurcations, and hysteresis in a large class of biological positive-feedback systems," *Proceedings of the National Academy of Sciences*, vol. 101, no. 7, pp. 1822–1827, Feb. 2004.

[18] A. L. Hodgkin and A. F. Huxley, "A quantitative description of membrane current and its application to conduction and excitation in nerve," *The Journal of Physiology*, vol. 117, no. 4, pp. 500–544, 1952.

[19] G. Pugliese Carratelli and T. Burghi, "Zenodo/cdc26surrogategradientslearning: v0.2 updated release," 2026.

[20] D. E. Kirk, *Optimal Control Theory: An Introduction*, 1st ed. Englewood Cliffsm NJ: Prentice Hall, 1970.